

Kiwix English Best

Use Guide · Version v20200903a · April 2, 2026

Overview

Kiwix English Best is a Python script that automates the selection, downloading, and maintenance of the best available English-language ZIM files from download.kiwix.org. ZIM files are compressed offline archives used by the Kiwix reader to browse Wikipedia, Stack Exchange, Project Gutenberg, and hundreds of other resources without an internet connection.

The script:

- Crawls the Kiwix server to find the best (newest, highest-quality) English ZIM file for each content group
- Downloads files using BitTorrent (via libtorrent) where available, with HTTP as a fallback
- Supports resuming interrupted downloads
- Verifies file integrity after download
- Detects and downloads updates when newer versions are available on the server
- Cleans up older duplicate versions to save disk space

Requirements

Requirement	Notes
Python 3.8+	Required
requests	<code>pip install requests</code>
tqdm	<code>pip install tqdm</code> (optional — progress bars)
libtorrent	Optional — enables torrent downloads
zimcheck	Optional — enables deep ZIM integrity checking (HTTP only)

Install Python dependencies:

```
pip install requests tqdm
```

To enable torrent support on Debian/Ubuntu:

```
sudo apt install python3-libtorrent
```

Getting Started

Run the script interactively:

```
python3 kiwix_english_best.py
```

You will see the main menu:

```
Kiwix English ZIM Tool v20260403a
1) Generate / download new best English ZIMs list
2) Download / resume from existing best English ZIMs list
3) Check for updates (new and newer ZIMs; includes cleanup)
4) Cleanup ZIMs directory (remove older duplicate versions)
Choose (1/2/3/4):
```

All ZIM files are stored in the zims/ subdirectory relative to the script's location.

Menu Options

Option 1 — Generate / Download New Best English ZIMs List

When to use: First-time setup, or when you want to refresh the full best-file selection from the server.

What it does

1. Crawls the Kiwix server to discover all available English ZIM files
2. Selects the best (newest date, then largest) file per content group, excluding unwanted variants (nopic, mini, lcc-*, etc.)
3. Saves the complete best list to kiwix_english_best.txt — always, even if all files are already present on disk
4. Checks which files from the list are not yet downloaded
5. If files are missing, shows a space check and prompts to download

Prompts

Prompt	Description
Download now? (y/n)	Whether to start downloading the missing files
Verbose output? (y/n)	Verbose shows per-file progress lines; non-verbose shows a compact updating display

■ The list file (kiwix_english_best.txt) is always saved with the full best selection from the server, regardless of what is already on disk.

Option 2 — Download / Resume from Existing List

When to use: Resume a previously interrupted download session, or re-download missing files from an existing list.

What it does

1. Checks whether `kiwix_resume_update.txt` exists (left by a failed Option 3 session) — if so, offers to resume those incomplete update downloads first
2. Otherwise loads `kiwix_english_best.txt`
3. Filters out files already present on disk (using a size-tolerance check)
4. Shows a space check for remaining files
5. Downloads any incomplete or missing files, resuming `.partial` files where they exist

■ If a download was interrupted, simply re-run Option 2. The script automatically detects `.partial` files and resumes from where it left off.

Option 3 — Check for Updates

When to use: Periodically, to download newer versions of ZIM files and pick up newly available content groups.

What it does

1. Checks for an incomplete previous update session — if `kiwix_resume_update.txt` exists, offers to resume it or run a fresh check
2. Crawls the server and compares the best available version of each group against what is on disk (using content date from filenames)
3. Reports updates (newer versions of files you have) and new groups (content you don't have yet)
4. Checks for torrent availability for each file to download
5. Downloads with per-file cleanup of the old version immediately after each new file completes verification
6. Saves a resume file (`kiwix_resume_update.txt`) if any downloads fail

Prompts

Prompt	Description
r) Resume / n) Check for all updates	Only shown if <code>kiwix_resume_update.txt</code> exists
Download updates? (y/n)	Whether to start downloading
Verbose output? (y/n)	Progress display mode

■ Option 3 uses torrent downloads where available, falling back to HTTP. Cleanup of the old version happens immediately after each file is verified.

Option 4 — Cleanup ZIMs Directory

When to use: If you have accumulated older duplicate versions of ZIM files.

What it does

1. Scans zims/ for files belonging to the same content group
2. For each group with multiple versions, shows which file will be kept (newest date) and which will be deleted
3. Shows the total space that will be freed
4. Asks for confirmation before deleting anything

Example output:

```
Duplicate groups found: 2
Group 'wikipedia en all maxi':
Keep : wikipedia en all maxi 2026-02 zim
Delete : wikipedia en all maxi 2026-02 zim (115 MB GiB)
2 file(s) will be deleted freeing 115 MB GiB
Proceed with deletion? (y/n):
```

■ Answering 'n' cancels without touching any files.

Non-Interactive Update Mode (Cron Jobs)

The script supports a fully non-interactive update mode suitable for automated scheduling via cron or any task scheduler.

Usage

```
python3 kiwix_english_best.py -u
# or
python3 kiwix_english_best.py --update
```

Behaviour

- Exits silently (exit code 0) if the zims/ directory does not exist — prevents cron failure emails when the storage volume is not mounted
- If kiwix_resume_update.txt exists, resumes the incomplete session
- Otherwise crawls the server and downloads all updates and new groups
- Non-verbose output — clean for log files
- Exits with code 1 if there is insufficient disk space or the server crawl fails

Example Cron Entry

To run every Sunday at 3:00 AM and log output:

```
0 3 * * 0 python3 /path/to/kiwix_english_best.py -u >> /path/to/kiwix_cron.log 2>&1
```

■ If your ZIM files are on an encrypted volume (e.g. VeraCrypt), the volume must already be mounted when the cron job runs. The script will exit gracefully if it is not.

File Reference

File	Description
zims/	Directory where all ZIM files are stored
zims/*.zim	Completed, verified ZIM files
zims/*.zim.partial	Incomplete downloads — safe to leave; will be resumed automatically
zims/*.zim.fastresume	Torrent resume data — deleted automatically on completion
zims/corrupt/	Files that failed integrity verification
kiwix_english_best.txt	Full best-file list generated by Option 1
kiwix_resume_update.txt	Incomplete update session resume file — written by Option 3, deleted on full completion
kiwix_download_failures.log	Log of all download failures with timestamps and error details

List File Format

```
# Format: filename|size_human|size_bytes|url|torrent_url
wikipedia_en_all_maxi_2026-03.zim|115.00
GiB|123456789|https://...|https://...torrent
```

Download Methods

Torrent (Preferred)

When a .torrent file is available on the Kiwix server, the script downloads via BitTorrent using libtorrent. This is faster and more reliable for large files, distributing the load across multiple peers. Torrent downloads are piece-verified by libtorrent, so only a ZIM header check is performed after completion.

HTTP (Fallback)

If no torrent is available, or if the torrent download fails, the script falls back to direct HTTP download from the Kiwix mirrors. Resume is supported via HTTP Range requests — interrupted downloads pick up from the byte offset already on disk.

Integrity Verification

Download Method	Verification
Torrent	libtorrent piece-verification during download; ZIM magic header check on completion
HTTP	ZIM magic header check; zimcheck -C -D (structure + directory) if zimcheck is installed

■ The -I flag (index check) is intentionally omitted from zimcheck for performance. Manual check:
 zimcheck -I yourfile.zim

Files that fail verification are moved to zims/corrupt/ and logged to kiwix_download_failures.log.

Tips and Troubleshooting

A download failed — how do I resume it?

Run Option 3. If a resume file exists, you'll be offered the choice to resume the incomplete session. The .partial file in zims/ will be picked up automatically.

The server crawl is very slow

The script scans each subdirectory of download.kiwix.org/zim/ in sequence. This is normal — the server has hundreds of directories. The scan typically takes 1–3 minutes.

I see "Found 0 update(s)" but I know there are new files

Check that your on-disk files have dates in their filenames (e.g. _2026-02). Option 3 compares content dates parsed from filenames. Files without a recognisable date pattern will not be compared correctly.

Torrent downloads show 0 seeds

Wait a minute or two — peers are discovered gradually. If seeds remain at 0 for several minutes, the torrent may be poorly seeded. The script will fall back to HTTP if the torrent stalls.

A file ended up in zims/corrupt/

The file failed integrity verification. Run Option 3 — it will detect that the completed .zim is missing for that group and offer it as an update. Alternatively, delete the corrupt file manually and run Option 2 to re-download it.

How do I see what files are scheduled for download?

Open kiwix_english_best.txt in any text editor. Each line lists a filename, its human-readable size, byte size, HTTP URL, and torrent URL.